CSMVessel0000000005 Cruise Ship Resilience_ Safety Systems

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 **Category:** Maritime / Safety

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Revision Drivers (all V2.0 documents)

1. **2025-2026 experimental and regulatory data** supersedes V1.0 specifications
2. **Solar Cycle 25 SSN ~137 peak** (observed) — elevated urgency for all Carrington-focused work
3. **Standards updates** — AASHTO 2024, IMO 2025, IEEE P2945 draft, DNV GL EMP-2, ClassNK FRP

V1.0 vs V2.0 Parameter Changes

Parameter	V1.0	V2.0	Consequence if Not Updated
SOLAS emergency power	Two-source (diesel + battery)	Three-source rule (SOLAS 2025): + LiFePO4 BFRP/MXene enclosed	SOLAS 2025 amendment creates compliance gap in V1.0 proposals
Lifeboat davit motor	Electric copper-wound winch	KNbO3-BaTiO3 ultrasonic motor + UHMWPE rope + ZrO2 blocks	GIC failure of electric winch prevents lifeboat launch — life safety critical
Fire detection	Conventional ionization sensors	Distributed optical fiber DTS temperature sensing (all-dielectric)	Metallic fire detection wiring fails in GIC — detection loss during likely concurrent fire scenario
GMDSS backup	DSC + GPS	Iridium Certus + PMMA optical fiber antenna feed (GPS-free)	GPS-dependent GMDSS unavailable during solar particle event that accompanies CME
Optical enclosure SE	Copper shielded box SE=45 dB	MXene Ti3C2Tx SE=92 dB for all SOLAS electronics	Life-safety electronics inadequately shielded in V1.0

Unchanged Elements

All core design philosophy, threat physics (GIC induction equations), material selection rationale, and fundamental architecture are carried forward unchanged from V1.0. Only the parameters listed above were revised.

End Versioning Delta | CSMVessel0000000005 Cruise Ship Resilience_ Safety Systems | June 2026